

# Measurement Notebook

Student sheet

Name: \_\_\_\_\_

Date: \_\_\_\_\_

**Safety:** In corridors and on stairs, walk slowly and do not look at the screen; hold the box with both hands.

## 1. What I see on the screen

At the top: the **Quantity:** list, empty for the first few seconds; **Temperature** comes first and is chosen from the start. Beside it, the label field; **Enter** saves it, the same as the green **Save the value** button. The big blue number is the live value, **not yet saved**. Below: **Delete last**, **Clear all** (asks first, then cannot be undone!) and **Export data**. The chart shows only the points of the chosen quantity, and its axis does not start at zero: **read the numbers on the axis**. **One quantity per map:** every point remembers its quantity (the **Quantity** column), but a map compares only like with like. **Wait 60 s at every stop** (90 s for CO<sub>2</sub>, 5 s for light), stand still, hold the box by its edges and do not breathe towards it. **Export before closing the window**.

## 2. What I do on my own

**Protocol** (circle one): **1** heat map / **2** light map / **3** CO<sub>2</sub> across spaces      **Box:** \_\_\_\_\_

**Quantity chosen:** \_\_\_\_\_ **Unit:** \_\_\_\_\_ **Height:** \_\_\_\_\_ m **Start time:** \_\_\_\_\_

**The exported file will be called:** \_\_\_\_\_

**First I predict. I do not change my prediction after I measure.** The highest value will be at \_\_\_\_\_, the lowest at \_\_\_\_\_, and the difference between them will be \_\_\_\_\_, because \_\_\_\_\_

**The labels. I copy them from here, letter by letter.** Rules: (1) lower-case letters only; (2) no accented letters (naïve, not naïve); (3) a hyphen, not a space; (4) at most three words; (5) only from the list; (6) **never leave the field empty** — the app writes #1, #2 on its own, and those can no longer be grouped.

1. **Heat map** (8 stops, **Temperature**): 1 door · 2 front-left-corner · 3 teacher-desk · 4 front-right-corner · 5 centre · 6 window · 7 radiator · 8 back-corner

2. **Light map** (9 stops, **Illuminance**): 1 desk-1-window · 2 desk-1-middle · 3 desk-1-door · 4 desk-3-window · 5 desk-3-middle · 6 desk-3-door · 7 desk-5-window · 8 desk-5-middle · 9 desk-5-door

3. **CO<sub>2</sub> across spaces** (7 stops, **CO<sub>2</sub>**): 1 outside · 2 empty-room · 3 occupied-room · 4 corridor-break · 5 library · 6 sports-hall · 7 outside-return

**Now I measure.** Row  $n$  is stop  $n$  from the list above. **The circuit is walked twice**, in the same order.

| # | Label | Time | Circ. 1 | Circ. 2 | Conditions |
|---|-------|------|---------|---------|------------|
| 1 |       |      |         |         |            |
| 2 |       |      |         |         |            |
| 3 |       |      |         |         |            |
| 4 |       |      |         |         |            |
| 5 |       |      |         |         |            |
| 6 |       |      |         |         |            |
| 7 |       |      |         |         |            |
| 8 |       |      |         |         |            |
| 9 |       |      |         |         |            |

**The threshold:** how big a difference has to be before I believe it.

**Spread** = the biggest Circ. 2 – Circ. 1 difference at the same place: \_\_\_\_\_ **Drift** = stop 1: \_\_\_\_\_ – \_\_\_\_\_ = \_\_\_\_\_

**Threshold** = spread + drift; e.g.  $0.3 + 0.5 = 0.8$ . Mine: \_\_\_\_\_ + \_\_\_\_\_ = \_\_\_\_\_

Highest: \_\_\_\_\_ Lowest: \_\_\_\_\_ Difference: \_\_\_\_\_, which ☐ **is** ☐ **is not** bigger than the threshold. *“Not detected” does not mean “does not exist”.*

### 3. The conclusion I write

I predicted that, and the data show that \_\_\_\_\_

The difference I trust is, because \_\_\_\_\_

For someone else to repeat our measurement, we would need to give them \_\_\_\_\_

#### Exit ticket

One thing that surprised you today, and one thing you would do differently if you repeated the measurement.

\_\_\_\_\_